

The eCommerce revolution

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This paper looks ahead to the mid-term and longer-term effects of the eCommerce revolution. We are in the last few years of the domination of the industrial economy — soon the information economy, or network economy, will take its place as the main engine of wealth. The underlying trends that are driving this revolution are discussed and guidelines offered for success in the new commercial world. Often these guidelines seem counter-intuitive and against conventional commercial wisdom.

The paper describes how customer and business relationships are being re-formed by Internet technologies, why the revolution is an all-pervasive irresistible force, and how it threatens previously profitable industries, while creating a multitude of new ones.

Perhaps the most fundamental change for companies will be the advent of supply networks, an alternative to traditional, linear supply chains. For consumers the most fundamental change will be the increase in their buying power. The new role of advertisers, the importance of trust, the threat from 'Big Brother', and the changes in the global economy are all examined.

There are four main sections that look at issues for companies, customers, network-economy trends, and finally, a section that describes why some companies are at risk.

1. Introduction — eLife on earth

Our current forms of commerce are very much a function of yesterday's communications techniques. Companies have built relationships with suppliers, partners and customers using slow, non-interactive and non-automated methods. Companies are now beginning to use 'eCommerce' and are finding ways to integrate new media and on-line communications into their business processes, and even to create entirely new business models that exploit the speed, interactivity and automation of the Internet. Radically new commercial structures are forming.

Until recently the best analogies for commercial activities have been strategies from war or sport. In the new commercial environment biology offers the best insights. The first life forms to evolve on Earth were simple, single cells. It took millions of years for multi-cellular life to appear. These life forms were limited as the cells needed to touch each other to co-ordinate — only basic structures were possible. However, with the advent of neurons and nerves, cells could communicate over distance and the diversity of life on earth exploded.

'Silicon chips linked into high-bandwidth channels are the neurons of our culture. Until this moment, our economy has been in the multi-cellular stage. Our industrial age has required each customer or company

to almost physically touch one another. Our firms and organisations resembled blobs. Now, by the enabling invention of silicon and glass neurons, a million new forms are possible. Boom!' [1]

This period of history is known as the Cambrian Explosion and it gives us many other clues as to the way eCommerce may move forward. There was very little stability, life forms and the environment were in constant flux, and creatures were prone to sudden extinction as their environment changed. This persuasive analogy indicates that we are at the beginning of what will be a fundamental revolutionary change to commerce.

There are many definitions of eCommerce but as eCommerce moves beyond a research idea into the market, and eventually grows to envelop almost all commercial activity, it can be seen that its primary difference is in the way it connects the participants. It can be loosely defined as innovation in two areas — at the customer interface and in the back-end systems which manage a company's internal operations and supply relationships. Innovation at the customer interface will involve extending the reach of merchants to consumers, and more importantly, will involve an enriching of the interactions between these parties. In the back-office, innovation will focus on reducing costs

associated with serving customers through creating demand-driven systems.

What will be affected by this commercial revolution? This paper will examine the effects that are predicted to occur in business, society, government, and our personal lives. Much of what is explored here is based on prototype businesses and trends that are already observable in the market — these trends are simply being extended to their next stages.

2. eCompanies

Companies can interact with their customers, partners, suppliers and employees instantly and globally — and, as a significant proportion of commercial output becomes digital, companies will be freed from the constraints of the physical world. The possibilities for commercial innovation seem almost endless. A range of significant trends that highlight novel ways in which companies can be organised are covered in this section.

2.1 Supply networks

Just as networking stand-alone PCs transformed business processes within organisations, networking stand-alone companies will transform business processes between organisations. Tighter relationships between organisations will be built using extranets. The ability to exchange information in this way will revolutionise how organisations interact with each other.

Initially this networking will lead to more efficient automation of supply chains. However, as the technology becomes more pervasive it will permit organisations to work in a much more collaborative way with suppliers and customers, and also with organisations that lie outside the traditional supply chain. This will lead to the formation of supply networks where not only is an organisation sharing information and working collaboratively with its immediate suppliers and customers, but also with a much wider range of organisations [2]. The Internet portal sites, such as Excite and Yahoo, are early examples of this mode of commerce [3, 4]. Many of the functions of these sites (i.e. stock quotes, travel, classifieds, dynamic maps, etc) are provided by a network of partner firms.

Existing supply chains evolved from the need to increase business efficiency. It is not practical for an individual to order a single tin of beans directly from the manufacturer. The cost to the manufacturer to process each order, deal with returns, ship the item and arrange billing for individual customers led to roles for intermediaries, such as distributors, stock holders and retailers — smaller, individual orders are aggregated into larger orders. This is changing with the ease and low cost of communications provided by the use of the Internet. For some goods the

classic supply chain is already being bypassed. One of the earliest Internet successes was Virtual Vineyards, a wine mail-order group. By organising many growers on its Web site and utilising efficient logistics, Virtual Vineyards has created a simple supply network [5].

As both experience and knowledge of the new ways of doing business grow, the relationships and the electronic ‘bonding’ between the organisations involved in supply networks will grow stronger and it will become increasingly difficult to draw boundaries between these organisations. A stage will have been reached where rather than having companies competing against each other, there will be supply networks competing against each other.

2.2 All companies will become eCompanies

Electronic firms will be highly efficient and will steadily outgrow, envelop or suddenly obviate the need for old-fashioned firms. Clearly Amazon.com is an eBusiness, but it relies on the very real businesses of post, printing and paper pulp production. Will these industries succumb to the eCommerce revolution?

Consider logistics, the science of efficiently moving goods and supplies — this requires effective use of the logistics company’s resources, e.g. maximally using load capacity, and ensuring vehicles are full on return journeys. The modern logistics firm uses global positioning systems (GPS) to check on its fleet, real-time parcel tracking, dynamic vehicle routing and traffic congestion warning systems to ensure maximum efficiency. With open electronic interfaces to its customers (e.g. mail order firms) logistics organisations become an easy-to-integrate component of the eCommerce infrastructure.

Consider agriculture, the oldest industry — it, too, is becoming wired. The largest and most modern farms use satellite positioning to track ‘intelligent’ machinery to allow the farm’s data systems to collect statistics on the spraying and yield for any particular area of land. The farm is linked to weather reports, crop pricing data, agriculture policy and advice information, and delivery schedules at grain processing plants, etc [6].

In this way all industry and human endeavour will become entwined in the network of eCommerce.

2.3 The virtual corporation

The concept of the virtual corporation has been envisaged for some time, but now it is a reality. Virtual corporations are dynamic and can be configured at will. The resources of the firm, whether people or partner organisations, can be called on when needed. They lay outside the small management core — they comprise the ‘exo-firm’ [7]. Many consultancy companies are organised

in this way. To fulfil a client's task the appropriate experts are assembled into a custom-company. If the client demands a lot of specialist expertise or a presence in many countries, it can be recruited. The appropriate people are found, they execute the task and then disband when it is finished.

There is none of the traditional resistance to change found in highly structured, hierarchical companies. In a hierarchy it is assumed information is scarce and therefore members are ranked so that orders are followed. Virtual corporations are networks and information flows freely from peer to peer. A virtual corporation relies entirely on communications and the kind of *ad hoc* organisation that the Internet allows. This organisation is all about change.

2.4 *eCommerce processes as reusable 'commerce objects'*

Companies' eCommerce services can be thought of as objects encapsulating certain functionality. These objects use the Web as an open interface and can be reused by others, in a modular way, to form more sophisticated, higher value services.

One example of this reuse, that was pioneered by Anderson Consulting, was BargainFinder [8]. BargainFinder is a comparison shopper (or 'shopbot') for CDs. A user enters the desired CD title, into just one Web form, and BargainFinder reuses the search functionality from many on-line CD stores to present the user with a list of comparative prices. The shopper no longer has to visit many CD stores to compare prices. In this case some on-line CD stores felt threatened and they denied BargainFinder access to their site. They were worried that shoppers would then shop on price alone, and that if this trend continued no human shoppers would visit their sites, only shopbots. Now, most on-line stores welcome comparison shopbots and understand how to co-exist with services that appear to 'overlay' them. Stores offer a few very good deals, which when found using shopbots, attract customers. Once there, the bargain hunters can be turned into loyal customers.

Other examples of commerce objects are essential parts of the buying process. Logistics, catalogue presentation [9], order taking, finance, profile management and data mining services are offered by firms to be reused by others. This allows complete on-line businesses, with all the necessary features, to be built simply by connecting and configuring readily available commercial building blocks.

This evolutionary trend, of absorbing and then extending existing entities, is typical of technological progress. This is complimented by the trend for finely focused niche services and personalisation. Where will this trend lead eCommerce? One possible outcome is that individuals will be able to construct their own fully

customised purchasing environments from the high-level tools and services provided on the Internet. A personal transaction-space containing real-time status reports of all of the things we buy and consume, help and advice from other buyers, loyalty programmes and promotions, which will then all integrate with our personal banking applications.

2.5 *Customers become employees*

As information technology becomes part of our every day life we notice that we can now do many tasks that previously required specialists. The obvious example is desktop publishing. We now create our own stationery, newsletters and even greetings cards. Often we will do the design part (personalisation) and let the specialist produce the goods. Technology gives us the tools to design our own products.

This trend continues on-line. Dell's 'Buy a Dell' site lets customers design a machine to their own precise specification [10]. Once the new computer is designed (i.e. amount of memory, hard disk size and other key features are chosen) the customer hits the 'buy' button and the component supply, assembly and logistics functions are invoked to supply the product in a just-in-time manner. No employees are needed to talk to the customer or to enter the customer or product details, and this reduces Dell's costs. As we take responsibility for more and more roles in the buying process, the distinction between employee and customer becomes blurred.

2.6 *Reaching niches*

The Internet allows a form of communication and co-ordination that was previously impossible or uneconomic — in marketing terms this was termed the 'hard middle'. The Internet allows mid-sized groups to communicate or be communicated with.

In the past, if one needed to communicate with a few thousand people who were not in one place, then it was very difficult. We have TV or radio to communicate with millions and letters to communicate with individuals. The Internet opens up the middle ground where current media struggles. Many of the innovations on the Internet exploit this. Niche marketing has come of age.

3. **eCustomers**

The eCommerce revolution is changing the customer's relationship with a company. Customers become more powerful as a result of sharing information [11]. This trend is then amplified by progressive new organisations who, in an attempt to win customers and loyalty, pander to these powerful customer needs.

3.1 *Buying communities*

A new form of intermediary is emerging to help to organise buyers in order to increase their buying power. They are often called 'virtual communities' and they act as a broker between buyers and products [12]. These communities act as the 'champion of the customer' helping the customer to get the best deal, and they provide merchants with access to informed customers who are eager to buy. They operate in particular niches, and in many ways are the natural, on-line manifestation of the traditional magazine. Advice and discussion forums are offered to ensure customers fully understand the product domain, and are certain about a buying decision.

To make a purchase we must first identify a need, then we analyse products and merchants, we negotiate the best deal, and finally we review our purchase. Buying communities understand and augment this buying behaviour [13].

When product domains are complex and marketing messages from manufacturers are confusing, then a buying community site is very valuable. They can offer the buyer help to understand and choose the most suitable product [14], they can compare merchants' prices [15], they can offer a potential customer reviews and feedback from previous customers [16], who are just like themselves [17], and they can even tell the customer the precise profit margin the merchant is making [18]. With such powerful advice customers will be sure of a purchasing decision and can make informed purchases with confidence. A good example of a site that offers all of these tools is DCresource.com, a buyers' advice site for digital cameras [19].

The key to making this service work is to instil a culture of feedback and review among customers. For a short period, when Amazon began, they offered \$1500 per day for the best book review that was submitted. This information 'primed' their recommendation system. It now contains enough review content to persuade customers that they will in fact like a book, and other related books, because similar people to them like these books.

This trend has threatening implications for big brands. In general, we resent exaggerated and confusing marketing messages but we often buy these products because we cannot find better, impartial advice. Big brands reduce buyers' doubt but not as well as buying communities can. We will expect to see products compared honestly, and the complaining letters as well as the glowing testimonials. Big brands cannot work like this — it does not seem to make commercial sense to fairly compare a competitor's product on one's own Web site. Big brands will lose customers whose loyalty will be won by impartial buying communities that they can trust.

3.2 *The customer is king*

It is in a company's interests to encourage their customers to talk to each other. By offering discussion facilities, customers will exhibit a smarter, more well-informed and collective intelligence. What these customers consume is then often produced by themselves — the 'audience are the authors'. This group of customers can help the company push its technology further, finding ever more creative ways to use the company's products [20]. They can help each other to solve problems and can provide early problem reports. Software companies have long used enthusiastic users like this to do beta testing (see the many Microsoft news groups dedicated to individual Microsoft products).

These users are often the most knowledgeable (often more knowledgeable than a company's own staff), and the resources these customers create can form the equivalent of on-line user guides and helpdesks.

Not only do these customers help existing customers but they will create new ones by enthusiastically advocating a product or service. By providing the means for customers to organise in this way a company can turn a previously disparate set of individuals into an easy-to-reach, loyal, niche market.

3.3 *Power to the purchaser*

eCommerce is shifting the balance of power during a transaction towards the customer. By using the Internet to help them form organised groups, customers are effectively creating 'Chambers of Commerce for Customers'. By sharing information, experience and details of previous transactions, customers can increase their buying power by acting as a single, larger group. In the longer term, it is likely that people's agents will work together to make large buying communities. A small piece of software could link together millions of people into a single market.

Traditionally a customer can have little influence on a firm. If a firm is powerful then a complaining letter or call is unlikely to cause any great effect — each customer is a lone voice. Even e-mail, though much easier to send than a letter, can simply end up in the electronic waste basket or get a worthless automatically-generated response.

Public discussion groups change the balance. A single customer's views can now be seen by all. This vastly increases a company's incentive to act (rather like the BBC's Watchdog programme does). It is now common for customers to search for news articles about a firm and its products. Customers can see what others think before deciding to buy — it is like having hindsight in advance! Dejanews, the news group indexer and search engine, makes this very easy [21]. There are many news groups dedicated to particular companies and products, specifically

to allow customers to pass on their good and bad experiences (see the news group 'uk.telecom' to find out what customers really think of BT and its products).

This is such a powerful means to regulate companies' behaviour, and to inform consumers, that it has been suggested that public discussion forums are made mandatory, and are managed by government watchdog organisations [22].

4. eTrends

4.1 *Commerce shapes the medium*

The nature of advertising will change. Broadcast advertising will be superseded by a type of new-media marketing where individual promotional efforts can be precisely measured in terms of success and increased revenues. Traditional advertising is a very imprecise activity. It is difficult to quantify the extra sales that advertising stimulates. There are survey techniques to estimate a campaign's effectiveness but there is no way for a company to measure the exact return on investment of any single promotional effort, or what exactly stimulated a buyer to make a purchase.

By contrast the Web affords many ways to measure an advert's effectiveness. When a buyer hits the 'buy' button, the advertisement, advertorial, special-offer, or any other advertising-oriented on-line media that brought the buyer to the purchasing decision, can be known. The power of any advert to induce a sale will be measurable almost immediately. Advertisers will spend their promotional resources only in the places where results are good. Advertising costs will not be fixed but will be performance-based.

How might a company manoeuvre into a position to exploit precise targeting and performance-based pricing? This is a role for yet another intermediary (or aggregation of intermediaries). We will call this function a Web advertising and promotions intermediary (WAP).

Customers are becoming increasingly aware that they are being tracked, logged and watched by companies who are eager to learn about them. Learning about customers makes it easier and cheaper for firms to sell. Therefore, customer profile information has a value. High quality, accurate data (given willingly by the customer) will become an extremely valuable commodity. Consider the loyalty card schemes in the UK (e.g. Tesco's ClubCard) which reward their customers, with discounts, to surrender precise information about shopping habits. Soon customers will demand some payback to give any valuable information about themselves. The WAP intermediary will manage these customer profiles and ensure that customers receive a reward for it.

Customers will share their buying histories, profiles, interests, reviews and feedback information with a WAP. Access to this information will be tightly controlled. Customers must be able to choose how, and with whom, the information is shared. The WAP will use this information to target customers on behalf of companies, charging only on success. As companies only pay for advertising based on success, they are able to make savings and can offer buyers better prices. The prices for the buyers will need to be genuinely and provably cheaper, and this saving is, in effect, the buyers' payment for surrendering their profiles. The first stages of this model have been pioneered by CyberGold [23].

New agencies and companies will form to serve this market and will threaten traditional big-media advertising firms (using TV, radio, print). The Web, and then interactive TV, will become powerful commercial media, defined largely by advertisers' needs (compare, for example, the TV soap operas of the 1950s). We will quickly move beyond banner adverts and the other old-fashioned techniques which have been hastily adapted for the Web.

4.2 *Trading in trust*

It has been argued that with a vast range of potential companies to deal with, customers will not be loyal. The same logic that predicted wholesale disintermediation also predicted that we would have many, shorter commercial relationships. However, in the digital world it is harder to build a trusting relationship, it takes time, but trust is a vital prerequisite for any eCommerce relationship.

Companies will need to work hard to build and keep their customers' trust. They will have to tell customers what information is being stored, what it will be used for, and with whom it will be shared. If customers are confident and happy with this they will begin to trust the company. Standards for describing privacy practices and rating schemes for companies' privacy procedures are appearing [24, 25]. Trust will be built over time, over the course of many transactions, and will have a mutual value to both company and customer.

Our privacy may be traded for goods or discount. In exchange for a year's free access to an on-line music site, customers may allow the music site to log their listening habits. They will receive junk mail as a result but they will have saved the subscription fees. Trust can also be an emergent property of a community of trading partners. After a transaction, both parties rate each other which then modifies their respective trust level. Over time a reliable rating level can be calculated. The auction market-place run by eBay uses such a system to allow members to judge the trustworthiness of potential trading partners based on their past transactions [26].

4.3 *Big brother*

For many years we have had concerns about 'Big Brother' and what information is known or can be deduced about us. Using the Internet exacerbates this. Everything we do can be stored and recalled later — our behaviour can be traced. A few decades ago we could be anonymous but now, with the advent of credit cards, mobile phones and the Web, organisations can tell what we have bought, where we have been, who we have spoken to and what we have seen.

Crimes are often solved with the aid of credit card records. Imagine how easy it will be to trace individuals when almost every interaction will leave an electronic record. It may be possible to analyse a person's entire lifetime. This would lead to many interesting possibilities in the future if the necessary historical data were available — history could be taught in schools by using data-mining techniques to create case studies of real people, and crimes could be solved many years after the event as data-mining techniques become more powerful.

Is the threat from Big Brother real? It seems unlikely that governments will have the will or wherewithal to do this. Instead the threat comes from businesses — the massive corporations that will have access to enough data. Our defence against such abuse will come from the enhanced customer power afforded by the Internet. Companies that transgress the new economy's protocols of trust and fair trade will be exposed and will be severely damaged.

4.4 *Free markets*

Information fuels the free market. The Internet has allowed thousands of entrepreneurs to find opportunities, such as large price disparities between countries. It has allowed them to deal with suppliers, buyers and shipping companies. Simply being able to send a digital image, in minutes, anywhere in the world, opens up countless new trading opportunities.

This will result in a levelling of prices around the world. Global prices will become more uniform (discounting import duty, government restrictions and shipping, of course). It is no coincidence that now, as the Internet pervades Europe, UK car prices tumble and a grey market is booming to exploit cheaper prices throughout the EC.

4.5 *Personal investments*

As the number of private investors increases, dealing becomes easier and cheaper, and as the quality of information rises, financial markets will be affected. The tools to help private investors are becoming very sophisticated. OpenSesame.com, an early experimental personal finance site, offered an entire portfolio

management site that tracked one's portfolio, made suggestions, gave advice, managed personal finances and planned for retirement. Other sites provide news and information, and a meeting space for private investors to share expertise [27].

With facilities like this it is expected that the proportion of private investments in the market will become truly significant, and the market may change in character [28]. Private investors do not trade as often as professionals — they are influenced by longer term growth and are less concerned with industry benchmarks and fashions. With more private investors the market should become more stable and share prices will be less prone to professional investors' whims.

A clear trend in the USA, for technology and knowledge companies, is that of offering share options in lieu of salary. This increases employees' commitment to their company and, potentially, their reward. This trend has paid off. Companies that have rewarded their employees in this way, have on average, returned 70% more value to their shareholders than those that did not [1].

The potential ramifications for the financial world are huge. This industry is comprised almost entirely of intermediation — organisations that mediate the flow of capital between those that have it and those that need it, taking a (large) commission in the process. The intermediation techniques are built from old-media processes and are inefficient. New players will redefine and re-intermediate financial services, and they will obviate the need for the old players. How far could this go? Consider insurance, which is a process of pooling risk. Now imagine an agent, running on your computer, that alerts you of insurance opportunities: 'Motoring opportunity: you must underwrite £3.17 (max); profit over a year = £0.17; actuary calculations by nSureUs.com. Do you accept this risk?' In this way the pooling of risk is among a group of individuals and is mediated by software agents. The risk assessment would still be done by a real actuary firm, but in some cases this could be automated too. The overheads and insurance commissions in this scenario would be a lot less than they are currently. Loan capital could flow through the networked economy in a similar, co-operative way.

4.6 *Connecting to commerce*

The stark changes we have seen so far in eCommerce have been built with the PC. Soon we will see mobile phones with colour screens and GPS locators [29]. Then the commercial innovation can extend to those who are not in front of their computers. The eCommerce revolution will become a mass-market phenomenon when the access technology presents no barriers — when we are not required to maintain unreliable PCs and dial-up connections.

With an on-line GPS device getting a cab will be a lot easier. We will select the 'call taxi' item from the menu, a request with your location goes to a taxi bureau, nearby drivers offer their best price and estimated arrival time, you pick the best offer and a cab arrives shortly.

All products can be connected. Minute chips, costing a few pence and with just a few functions can be attached to every-day items [1]. These chips will remember simple data and will respond when queried by radio or infra-red. Products could tell us how many times they had been used, how much they had cost, and where they were purchased. If we liked an item, for example, belonging to a friend, we could capture its details with our PDA and choose to have one delivered. Our friend may even get an automatic commission from the merchant's affiliate programme.

4.7 Globalisation

Clearly the Internet and the eCommerce revolution are global phenomena. Many geographic constraints will be removed. What changes can we expect to see?

Global mobile communications will cause a step change in the possibilities of regions that have had no previous form of telecommunications. There will be new clusters of service industries outside the traditional centres of intellect [30]. We have already seen the growth of the Indian and East European software industries. Next, perhaps, Third World citizens will be able to bypass their nation's poor communications and commercial infrastructure and take part in the global economy.

There are now global telephone networks which require global billing and global account administration. These trends will affect other industries too as a greater number of smaller companies become multinational, and customers become increasingly mobile. We will see people in one country, receiving services from another country, which are billed and administered from a third country. Customers will choose which companies to deal with, not based on their proximity, but rather on the reputation of their brand and their customer care.

Making cash digital will improve the seamlessness of international purchases. Even without a common official currency, electronic cash systems will create international *de facto* standards. This will reduce the currency exchange charges that we currently pay. This is often one of the justifications cited for a common European currency, but global electronic monetary union will happen by stealth.

Companies will choose 'countries of convenience' to locate their servers where tax laws are favourable. For people, too, the notion of residency becomes irrelevant. The technology exists to allow us to carry out transactions from almost any jurisdiction, and therefore the conventional 183-

day test, to determine where one pays taxes, becomes untenable [31]. With these threats to the revenues of conventional taxation systems, governments will need to act skillfully to maintain the size of their exchequers.

5. eThreats

Clearly the eCommerce revolution will lead to business casualties. This section outlines the characteristics that expose traditional businesses to danger from new players.

5.1 Threatened industries

Which industries are threatened by the revolution in eCommerce? As has been shown throughout this paper, eCommerce thrives on an incumbent's inefficiencies. Industries that are founded on an old style of intermediation, and also extract large margins from the transactions that they broker, are fertile regions in which new, highly efficient players will grow. Only businesses where the traditional or quaint nature of its product is a unique selling point can survive by not adapting.

Some industry sectors will undergo an entire restructuring. In the travel industry, commodity retailing and personal finance sectors, changes are already well under way. New players will attract customers with innovative and cheap services, and will use this body of customers to bypass the current dominant players [32]. These changes will be rapid and brutal. The growth rates will be spectacular. The classic eCommerce case study, Amazon.com, exemplifies this threat. In little over three years it had become a multi-billion dollar book, CD, video and drug retailing behemoth.

Other industry sectors may not face such a thorough restructuring, but the marketing, customer management and back-end processes of virtually every firm will be affected. If companies try to resist the inevitable tide of eCommerce, they will be submerged. In fact, if they do not seize the initiative and act quickly, even if this will threaten their existing profitable business, they may still go under — it is adapt or die.

5.2 Increasing returns

The industrial economy is characterised by diminishing returns — beyond a certain point, the cost of increasing a firm's capacity will exceed the profit gained from the increase. In contrast, the digital economy is characterised by increasing returns — the more you sell, the more you sell [33]. Increasing returns cause one player to grow, very rapidly, to dominate a market space. Microsoft was an early example of an increasing returns business. The more its products sold, the more likely it was that new customers would buy its products so that they could interoperate with existing users (however, dominant increasing returns

positions do not last forever — in many ways the open platform of the Internet weakens Microsoft's position). Amazon, too, is an increasing returns business. Its massive database of customer opinion and reviews means that customers are more likely to buy via Amazon because they will have the best chance of making a good, well-informed purchase.

Increasing returns cause the biggest to get bigger and to lock out all other competitors (which will increasingly arouse the attention of fair trade bodies). Understanding this growth dynamic is very important — it is vital to move fast and to grab market share as there is often no second place.

6. Conclusions

This paper has shown that new on-line and communications technologies have precipitated a revolution in commerce. This revolution is real and it will have far-reaching effects.

The young revolutionaries in this case are the entrepreneurial start-up CEOs, who, armed with potent new business models, and preaching attractive new consumer philosophies, are attracting customers in droves; but as in any revolution there will be fatalities. In this revolution it will be the old-fashioned, sluggish, inefficient firms who are up against the wall.

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References

- 1 Kelly K: 'New rules for the new economy', Forth Estate (1998).
- 2 Project ION — <http://www.labs.bt.com/people/callagjg/ion/supply.htm>
- 3 Excite — <http://www.excite.com>
- 4 Yahoo — <http://www.yahoo.com>
- 5 VirtualVineyards — <http://www.virtualvineyard.com>
- 6 Steward S: 'Future robotics', (available from author) (1992).
- 7 Murphy S: 'Corporate Metamorphosis: the effect of new media', First Monday (<http://131.193.153.231>), 1, No 1 (1996).
- 8 BargainFinder — <http://bf.cstar.ac.com/bf/>
- 9 iCat — <http://www.icat.com>
- 10 Dell — <http://www.dell.com/store/>
- 11 Barrett S: 'Fulton fish market case study (how different ethnic communities attain different prices from the same market as a result of their propensity to share buying experiences)', Internal case study, LBS (available from author) (1992).
- 12 Hagel J and Armstrong A G: 'Net Gain: expanding markets through virtual communities', HBS Press (1997).
- 13 Steward S: 'Intelligent on-line purchasing', British Telecommunications Eng J, 17, No 1, p 36 (1997).
- 14 PersonaLogic — <http://www.personalogic.com>
- 15 Acses — a shopbot for books — <http://www.acses.com>
- 16 Amazon — <http://www.amazon.com>
- 17 FireFly — <http://www.firefly.com>
- 18 Auto-by-Tel — <http://www.auto-by-tel.com>
- 19 DC Resource — <http://www.dcresource.com>
- 20 Dutta S: 'Success in Internet time — a BT global information exchange report', see — <http://www.infoexchange.bt.com> (1999).
- 21 DejaNews — <http://www.dejanews.com>
- 22 Andresen T: 'Consumer power via the Internet', First Monday (<http://131.193.153.231>), 4, No 1 (1999).
- 23 CyberGold — <http://www.cybergold.com>
- 24 Platform for Privacy Preferences — <http://www.w3.org/P3P/>
- 25 Website trust rating organisation — <http://www.truste.com/>
- 26 eBay — <http://www.ebay.com>
- 27 Motley Fool — <http://www.fool.com>
- 28 McRae H: 'The long view', Net Profit, No 22 (October 1998).
- 29 Symbian — <http://www.symbian.com>
- 30 McRae H: 'The long view', Net Profit, No 23 (November 1998).
- 31 Muscovitch Z: 'Taxation of the Internet', First Monday (<http://131.193.153.231>), 2, No 10 (1997).
- 32 PriceLine: on-line travel demand aggregation — <http://www.priceline.com>
- 33 Wired — <http://www.wired.com/wired/archive/3.10/arthur.html>



Simon Steward joined BT in 1984 as an apprentice at BT Laboratories. After his apprenticeship he joined the submarine optics division where he worked on laser amplifier repeater systems. He graduated from the University of Warwick in 1991 with a BEng in Electronic Engineering.

Since returning to BT Laboratories, he has worked in the cybernetics group where he studied self-organising systems, and recently in the electronic commerce research group investigating emerging on-line markets and trends.



James Callaghan joined BT in 1986, after graduating from Trinity College, Dublin, with a first-class honours degree in electrical engineering. He subsequently gained an MSc from the University of Essex, a Graduate Diploma in Marketing from the Chartered Institute of Marketing and an MBA from Henley Management College. His early work in BT was involved with all aspects of network management, ranging from mathematical modelling, hardware and software design, to network modelling. He also spent several years working on and leading large research projects in the European RACE programme. More recent work has focused on investigating the likely impact of new technologies on the way companies conduct their business. He is currently on secondment from BT Laboratories to the University of Cambridge where he is researching inter-organisational networks and is pursuing a PhD on the role and impact of intranets.



Tim Rea began working in eCommerce in 1996 and has been involved in a number of projects relating to the development and evolution of trust services.

His work has included a broad analysis of the nature of trust and of the roles that trust plays in electronic commerce and electronic transactions, and he has used this work to fuel the evolution of BT's portfolio of trust services. Other current work includes general eCommerce analysis and consultancy, a collaborative research project with NTT and a collaborative project with the

International Chambers of Commerce.

Tim holds a BEng from the University of Sheffield, an MSc from the University of London and is in the process of completing an MBA at London Business School.